

XIAOYU HE

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PROFILE

PhD researcher in Computer Science at the Czech Technical University in Prague, working in the Optimisation AI team under the supervision of Jakub Mareček and funded by the CoDiet Horizon Europe project. My research focuses on causal inference, learning dynamical systems, machine learning and biomedical data analysis, with particular interest in developing causal learning methods for high-dimensional biomedical and metabolomics data and applying them to real-world problems such as personalised nutrition recommendation systems.

PROFESSIONAL EXPERIENCE

Research Assistant, Czech Technical University in Prague (CVUT) Prague	Feb 2023 to Present
- Research at the intersection of optimization, causal inference, and machine learning. - Develop scalable methods for DAG-based causal discovery, including ExDBN, ExDAG, and Learning of Maximally Ancestral Graphs. - Improve solver efficiency using cliquewidth callback constraints. - Study non-smooth, non-convex optimisation for deep neural network training and complex dynamical systems in joint clustering settings. - Apply causal and machine learning methods to personalised nutrition by integrating phenotype and multi-omics data, including HEI, lipidomics, microbiome, and clinical biomarkers.	
Research Assistant, The First Affiliated Hospital of Xi'an Jiaotong University	Aug 2020 to Jan 2023
- Studied the gut microbiome, colorectal cancer, and related molecular mechanisms. - Contributed to multiple research projects, including Altered gut microbiome composition by appendectomy contributes to colorectal cancer. - Applied microbial source-tracking methods, including SourceTracker and FEAST, with preprocessing using QIIME 2, DADA2, and DEBLUR. - Performed trajectory inference and pseudotime analysis using Monocle2/3, psupertime, SCOUT, Slingshot, diffusion pseudotime, and RNA velocity. - Built a strong foundation in bioinformatics methodology, microbiome analysis, and complex biological data interpretation.	
Operation Data Analyst, Seagate International Technology (Wuxi) LLC	Dec 2017 to Jul 2020
- Developed and maintained monitoring dashboards for operational and engineering metrics, improving visibility into system performance and process status. - Performed statistical modelling, trend analysis, and anomaly detection on large-scale operational data to identify irregular patterns and potential risks. - Collaborated with engineering teams to deliver data-driven analyses, support troubleshooting, and inform process optimisation and performance improvement.	

EDUCATION

Czech Technical University in Prague PhD in Computer Science Expected completion: Dec 2026	Feb 2023 to Present
University of Manchester MSc in Statistics (Merit)	Sept 2016 to Jun 2017
University of Liverpool BSc in Mathematics	Sept 2014 to Jun 2016
Xi'an Jiaotong-Liverpool University BSc in Applied Mathematics	Sept 2012 to Jun 2014

AREAS OF EXPERTISE

- Python&R - Hydra and mlflow - statistical modelling	- Julia - Time-series modelling - Bayesian Networks	- MATLAB - causal inference - Optimisation methods	- Git - deep learning - multi-omics analysis
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LANGUAGES

English (Fluent), Mandarin (Native), Spanish (Basic), Czech (Basic)